The Mole

Exam code: 0620 & 0971

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Written by: Caroline Carroll
Reviewed by: Stewart Hird
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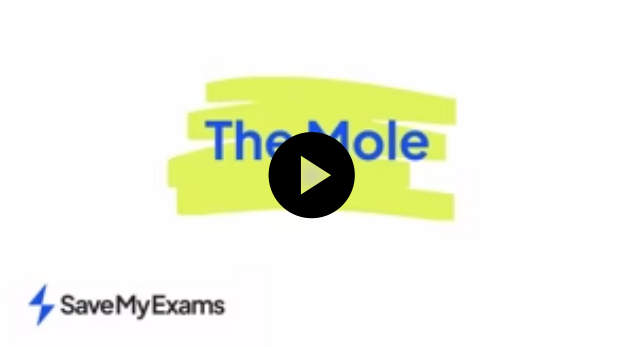

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The mole & the Avogadro Constant

Extended tier only

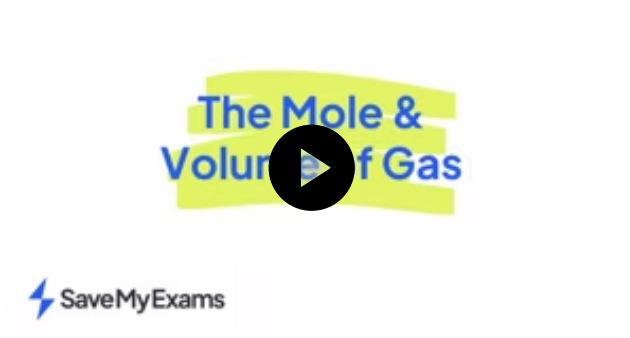
- Chemical amounts are measured in moles
- The mole, symbol mol, is the SI unit of **amount of substance**
- One mole of a substance contains the same number of the stated particles
  - This can be atoms, molecules or ions
- One mole contains **6.02×10^{23}** particles; this number is known as the **Avogadro Constant**
- For example:
  - One mole of sodium (Na) contains 6.02×10^{23} atoms of sodium
  - One mole of hydrogen (H_2) contains 6.02×10^{23} molecules of hydrogen
  - One mole of sodium chloride (NaCl) contains 6.02×10^{23} formula units of sodium chloride
- The mass of 1 mole of a substance is known as the **molar mass**
  - For an element, it is the same as the **relative atomic mass** written in grams

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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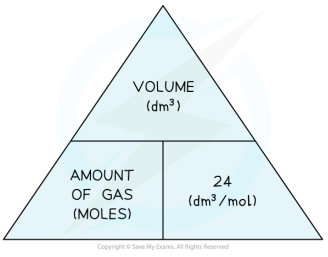
The mole & volume of gas

Extended tier only

Molar volumes of gas

- **Avogadro's Law** states that at the same **temperature** and **pressure**, equal amounts of gases occupy the **same volume** of space
  - e.g. 1 mole of hydrogen gas occupies the same volume as 1 mole of methane gas
- At room temperature and pressure, the volume occupied by one mole of any gas was found to be **24 dm^3** or **24,000 cm^3**
  - This is known as the **molar gas volume at RTP**
  - RTP stands for "room temperature and pressure" and the conditions are **20 °C** and **1 atmosphere** (atm)
- From the molar gas volume, the following formula triangles can be derived:

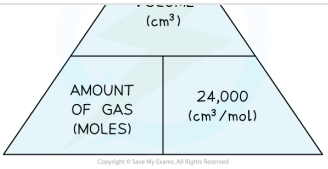
Molar gas volume (dm^3) formula triangle



This shows the relationship between moles of gas, volume in dm^3 and the molar volume

- If the volume is given in cm^3 instead of dm^3, then divide by 24,000 instead of 24:

Molar gas volume (cm^3) formula triangle



This shows the relationship between moles of gas, volume in cm^3 and the molar volume

- The formula can be used to calculate the number of moles of gases from a given volume or vice versa
  - Simply cover the one you want and the triangle tells you what to do
- For example, to find the volume of a gas:
  - **Volume = Moles x Molar Volume**

Examples of Converting Moles to Volume Table

Gas	Amount (moles)	Volume
Hydrogen	3	(3 x 24) = 72 dm^3 (3 x 24000) = 72000 cm^3
Carbon dioxide	0.25	(0.25 x 24) = 6 dm^3 (0.25 x 24000) = 6000 cm^3
Oxygen	5.4	(5.4 x 24) = 129.6 dm^3 (5.4 x 24000) = 129600 cm^3
Ammonia	0.02	(0.02 x 24) = 0.48 dm^3 (0.02 x 24000) = 480 cm^3

- For example, to find the number of moles of a gas:
  - **Moles = Volume ÷ Molar Volume**

Examples of Converting Volume to Moles Table

Gas	Volume	Moles
Methane	225.6 dm^3	(225.6 ÷ 24) = 9.4 mol
Carbon monoxide	7.2 dm^3	(7.2 ÷ 24) = 0.3 mol

Oxygen	1200 cm^3	(1200 ÷ 24000) = 0.05 mol

Examiner Tips and Tricks

- You are not expected to know the value of Avogadro's constant
- But, you do need to know the equation as well as how to use and re-arrange it

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